

When Do Taxpayers Want a Tax Inspector?

Personal Interaction, Private Concessions and Correction

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Abstract

Personal tax administration offers both private concessions and help with mistakes. I examine taxpayer choices using linked administrative records, independent measurements and a survey experiment with 2,100 Lahore owners. A model separates total payment, assessment state, assessor valuation and the net cost of a paid transaction. Across 21,001 paired tasks, 59.8% choose the satellite at equal correct bills, yet 58.5% of eligible initial satellite choosers subsequently accept a paid inspector offer. State dependence in transaction valuation varies across specifications; errors do not produce a stable additional preference for inspectors. Independent property evidence attracts substantial demand. Expected correctability is strongly associated with assessor choice. These findings distinguish preferences over administration from acceptance of particular transactions and describe the correction setting in which those choices arise.

JEL: H26, H71, D73, O33

1 Introduction

For many taxpayers, dealing with the State means dealing with someone who can change what they owe. Front-line officers, hence, become the first line of contact for taxpayers and perform multiple functions. A tax inspector can check a disputed record, correct a mistake and explain how to obtain an amendment. The same discretion can allow the inspector to offer a favourable bill in exchange for an informal payment. Personal interaction therefore combines an opportunity for assistance with an opportunity for a private deal. Distrust of officials does not reveal which of these functions taxpayers value, or when they are willing to transact with an official they would otherwise prefer to avoid. What do taxpayers value in personal tax administration?

Personal interaction matters because frontline officials are often where formal tax rules become actual liabilities (Mascagni et al., 2025; Dimitrov et al., 2026; Boll, 2015). The same discretion that allows an official to correct records and help taxpayers navigate administrative requirements can also enable negotiated departures from those rules, affecting who pays, how much they pay, and whether revenue reaches the state (Piracha and Moore, 2015, 2016; Raaphorst and Groeneveld, 2018). This tension matters in Pakistan, where weak revenue mobilisation and limited tax compliance make tax administration especially consequential (Hasan et al., 2024; Hasan and Sheikh, 2025). These concerns are visible in taxpayers' own perceptions. A recent survey of taxpayers from Lahore, the provincial capital of Punjab, Pakistan shows that 62.2% of the property-tax collection gap is attributed to informal payments, 37.4% report no trust at all in property-tax staff, and 64.3% disagree that Punjab uses tax revenue to provide goods and services for people like them. The underlying problem extends beyond Punjab. Across 34 African countries surveyed by Afrobarometer in 2019–2021, 35% of respondents believe most or all tax officials are corrupt, only 41% trust the tax office somewhat or a lot, and 61% find it difficult to establish what taxes or fees they owe (Isbell, 2022).

Such concerns give an impersonal administration an intuitive appeal: liability should depend on a rule rather than a relationship. Technology offers one route towards this objective by replacing discretionary face-to-face interactions with standardised procedures and verifiable information. Electronic filing, for example, can reduce opportunities for both extortion and collusion by limiting direct contact between taxpayers and officials (Okunogbe and Pouliquen, 2022), while third-party information can make tax enforcement less dependent on an individual official's judgment (Pomeranz, 2015). Recent work suggests that digital tools may also substitute for some forms of personalised assistance, including in property-tax disputes (Holz et al., 2026). Yet replacing the official can also remove access to local information and assistance that help taxpayers navigate the system (Balán et al., 2022). The case for reducing personal discretion therefore leaves an unresolved question about the services and financial advantages citizens seek from it.

I answer this question by formalising the problem as a taxpayer choosing among assessment arrangements that differ in the assessor, the initial assessment and the payment required. The model separates the value taxpayers place on the assessment of their tax liability, their prefer-

ences over the assessor and the value of entering a paid transaction. It therefore distinguishes a preference for impersonal administration from willingness to accept a favourable deal from an official.

I study urban property-tax administration in Lahore, the capital of Punjab, Pakistan's most populous province. Punjab provides an unusually useful setting because property-tax administration is moving from officer-mediated assessment towards increasingly rule-based and technological administration. Earlier GIS reforms reduced opportunities for inspectors to exercise discretion in assessment, while the 2024 reform shifted valuation from annual rental value to capital value based on administrative rates (World Bank, 2020; Government of Punjab, 2024). Lahore therefore allows me to study demand for personal administration precisely as the state is attempting to reduce it. Against this backdrop, I test these distinctions using linked administrative records and a survey experiment with 2,100 property owners in Lahore. The experiment uses a Bayesian adaptive choice design in which owners make repeated comparisons between assessment receipts generated either by an inspector or by a satellite-based system. By varying the assessment state, the availability of a paid concession and the resulting expenditure, I recover the trade-offs taxpayers make between assessor type, assessment accuracy and financial gain. The adaptive design concentrates information on the comparisons that identify whether taxpayers value impersonal administration itself, rather than simply the outcomes it delivers.

Three findings stand out. First, when both systems produce the correct bill at the same cost, 59.8% of owners choose satellite assessment. Second, this preference does not imply rejection of personal discretion: among eligible owners who initially choose the satellite, 58.5% subsequently accept a paid concession from an inspector. Third, choices also vary with whether taxpayers expect assessment errors to be correctable. Together, the results show that demand for impersonal administration can coexist with demand for the opportunities created by personal discretion. These findings imply that the case for impersonal administration is not simply a matter of replacing discretion with technology. Taxpayers may prefer technology while still valuing the corrective and financial opportunities created by personal discretion, so reform must confront both sides of that trade-off.

This paper contributes to the literature on tax administration by examining demand for personalised administration from the taxpayer's perspective. Existing work explains the organisation and persistence of personalised collection and shows how incentives facing tax collectors affect revenue mobilisation and reported bribery (Piracha and Moore, 2015, 2016; Khan et al., 2016). I instead study how taxpayers value distinct components of personal administration within a common choice framework: the identity of the assessor, the terms of the transaction and the availability of correction when an assessment is wrong. This distinction matters because dissatisfaction with existing fiscal institutions need not imply a preference for removing personal interaction. The experiment identifies which features of that interaction taxpayers seek to retain and which they prefer to replace.

The paper also contributes to the emerging literature on technology and discretion in tax administration. Knebelmann et al. (2026) compare bureaucratic and algorithmic property val-

uation, while Okunogbe and Pouliquen (2022) study how electronic filing changes compliance costs and interactions with tax officials. I complement this work by holding the financial consequences of assessment arrangements explicit and examining how taxpayers trade off impersonal assessment against the opportunities created by personal interaction. The results therefore speak to a central design problem in administrative reform: technologies that reduce discretion may also remove functions that taxpayers value, particularly access to correction. The experiment identifies stated preferences over these institutional arrangements rather than the revenue or welfare effects of introducing automated assessment.

The rest of the paper is structured as follows: Section 2 develops the problem using the baseline surveys and institutional setting. Section 3 presents the framework. Sections 4–6 describe the data, define outcomes and set out the empirical strategy. Section 7 answers the three research questions. Section 8 concludes.

2 Context

2.1 Study Site

The study site is Lahore, the provincial capital of the province of Punjab, Pakistan. With a population of 13 million people, Lahore is the world’s 15th largest city. For getting access to the administrative data, I partnered with Punjab’s Excise and Taxation (E&T) Department, a provincial government revenue authority that administers the collection and billing of the primary property tax in metropolitan cities. In Lahore, E&T administers the taxation of almost 1 million properties in Lahore.

2.2 Property-value Assessment and the Role of Front-line Tax Officers

Punjab’s Urban Immovable Property Tax (UIPT) is administered by the provincial Excise and Taxation Department. The department divides the urban jurisdictions subject to UIPT into a five-level geographical hierarchy, illustrated in Figure A.1. The primary unit for property inspection, assessment, collection, and enforcement is called the tax “circle”. Each circle is assigned an inspector, supported by a clerk and a constable. The inspector leads property visits, determines the relevant property characteristics, and takes primary responsibility for the recommended assessment. The clerk and constable assist with measurement, documentation, record preparation, and the delivery of notices. Accordingly, the term “officer” in the theoretical analysis refers to the inspector as the relevant front-line decision-maker.

The circle team cannot enter its recommendation directly into the official tax record. It must submit the assessment to the Excise and Taxation Officer (ETO), whose approval is mandatory. The ETO reviews the measurements, classifications, and supporting record prepared by the inspector-led team. The process therefore separates informational control from formal au-

thority: the inspector controls the production of property-specific information, while the ETO controls whether the recommendation receives formal approval. For analytical tractability, the model assumes that the ETO is not a party to any informal bargain between the inspector and the taxpayer. ETO approval nevertheless matters because the inspector must produce a record that is sufficiently complete and internally consistent to obtain formal approval.

2.3 The Taxpayer–Administration Relationship

A recent survey reveals the taxpayer– administration relationship in the study site. It distinguishes three related but conceptually separate dimensions of the fiscal relationship: perceived diversion during collection, trust in the institutions administering the tax and perceptions of the return from taxation. These distinctions matter because dissatisfaction with one dimension does not reveal which features of personal administration taxpayers would prefer to retain or remove.

Table 1 reports these measures. On average, citizens attribute 62.2% of the gap to informal payments, and 71.0% assign more than half of the gap to this explanation. These responses indicate a widespread perception that private transactions divert revenue from the formal collection process. They should not be interpreted as estimates of actual revenue loss or of the prevalence of bribery among tax officials.

Table 1: Baseline surveys: perceptions of collection, trust and public services

Measure	Count / respondents	Percent
<i>Panel A. Perceived diversion from collection</i>		
Mean collection gap attributed to informal payments	Mean / 4,897	62.2
Attribute more than half the gap to informal payments	3,476 / 4,897	71.0
<i>Panel B. Trust in officials, government and other citizens</i>		
No trust at all in property-tax staff	1,831 / 4,897	37.4
No trust at all in government	1,699 / 4,897	34.7
No trust at all in other Lahore citizens	738 / 4,897	15.1
<i>Panel C. Perceived return from taxation</i>		
Disagree that Punjab uses tax revenue for people like them	3,147 / 4,897	64.3

Notes: Citizen baseline survey from the same study site, Phase 1. The attribution question allocates the collection gap between informal payments and collection difficulties; its mean is a perceived share, not an audited revenue loss. Trust uses only the unambiguous no-trust endpoint. The service item combines strongly disagree and disagree to some extent. These are separate measures; no index is constructed.

Distrust extends beyond the collection process itself. No trust at all in property-tax staff is reported by 1,831 respondents, or 37.4%, while 1,699 respondents, or 34.7%, report no trust at all in government. By comparison, 15.1% report no trust at all in other Lahore citizens. The pattern therefore points to a particular concern with the fiscal relationship rather than a general unwillingness to trust others. A separate question captures perceptions of the fiscal

return: 3,147 respondents, or 64.3%, disagree that the Government of Punjab uses tax revenue to provide goods and services for people like them.

Similarly, a survey of front-line officers provided a complementary view of the same collection environment. Among 292 officials ranking five explanations for incomplete collection, weak penalties and poor public services receive the most first-place rankings. Taxpayers offering informal payments are ranked first by 9.6% of officials in the low-value property scenario and 8.2% in the medium- and high-value scenario. Because these questions differ from the citizen attribution measure, the responses are not quantitatively comparable. They nevertheless indicate that incomplete collection is perceived to reflect enforcement constraints and the fiscal exchange as well as private payments.

These baseline patterns establish citizen dissatisfaction with several dimensions of the existing tax relationship, but they do not identify what taxpayers would prefer instead. In particular, they cannot distinguish demand for predictable assessment from demand for lower payments, assistance in correcting errors or continued access to a particular official. This distinction is central because the officer may be both a source of discretion and a source of administrative assistance.

2.4 Personal Administration and the Move toward Automation

A taxpayer who distrusts the administration may nevertheless value access to an inspector when an assessment is wrong. Property-tax liability depends on recorded property characteristics as well as the valuation rule, so a correctly applied formula can still produce an incorrect bill. Front-line officials can inspect the property, examine supporting documents and initiate authorised corrections.

Such assistance is distinct from the financial advantages available through informal accommodation. Piracha and Moore (2015, 2016) describe how inspectors' local knowledge, continuing relationships with property owners and discretion over assessment and enforcement sustain arrangements that can benefit both parties while limiting public revenue. Personal administration can therefore provide access to lawful correction as well as private concessions. I distinguish *non-appeal administrative correction*, an authorised amendment within the assessment chain, from *formal appeal*, which follows the statutory appeal route. A *private concession*, by contrast, reduces the official demand in exchange for a private payment to the inspector. An administrative correction is not evidence of such a bargain.

This combination of assistance and discretion is central to understanding what a less personal administration would replace. Punjab's earlier GIS reforms sought to reduce inspectors' discretion, while the 2024 reform shifted valuation from annual rental value to capital value based on administrative rates (World Bank, 2020; Government of Punjab, 2024). My interviews with senior bureaucrats motivated the experimental comparison between inspector and satellite-based assessment. This comparison is distinct from the implemented valuation reform and does not presume that automation eliminates recourse. Rather, it examines taxpayers' stated choices

between assessment arrangements when both the assessment state and the financial terms of personal interaction are explicit. Section 3 formalises these trade-offs in a model that distinguishes preferences over the assessor from willingness to accept a private concession under different assessment states and payment terms.

3 The taxpayer's decision environment

A property owner receives a tax assessment and must decide how to settle the liability. The owner can pay the demand and may have access to an administrative route for correcting an error. An inspector may also offer a lower official payment in return for a bribe. Lawful correction and a paid concession are distinct. The model takes the inspector's assessment and any concession as given and describes the taxpayer's choice among the available terms.

3.1 Assessment, payments and payoffs

Let i denote the taxpayer and j a feasible way of settling the liability. The taxpayer recognises the initial assessment state, before any correction or concession:

$$e_{ij} \in \{0 \text{ (correct)}, O \text{ (overstated)}, U \text{ (understated)}\}.$$

The decision follows the communication of the final payment terms. Total expenditure is

$$\underbrace{p_{ij}}_{\text{total payment}} = \underbrace{T_{ij}}_{\text{final official payment}} + \underbrace{B_{ij}}_{\text{bribe}}, \quad D_{ij} = \mathbf{1}\{B_{ij} > 0\}.$$

Here $B_{ij} \geq 0$, and D_{ij} indicates whether a bribe is required. Payments and monetary valuations are in rupees. An available lawful correction can change T_{ij} without requiring $B_{ij} > 0$; the process of obtaining it is not separately modelled.

Compare human assessment, $S_{ij} = 0$, with prospective automation, $S_{ij} = 1$. Automation has $B_{ij} = D_{ij} = 0$. These are institutional comparisons, even when taxpayers cannot switch arrangements.

The three valuations beyond expenditure are:

Symbol	Meaning
$h_i(e)$	Value of the initial state under human assessment without a bribe; $h_i(0) = 0$.
$\beta_i(e)$	Value of automation relative to human assessment, at equal state and payment without a bribe.
$m_i(e)$	Additional transaction cost, beyond the bribe's cash cost in total payment.

These valuations can have either sign. In particular, $m_i(e) > 0$ denotes reluctance to enter the paid transaction; $m_i(e) < 0$ denotes attraction at equal total expenditure.

The same payoff function can be written separately for each arrangement:

$$\text{Human assessment: } V_{ij} = \begin{cases} -p_{ij} + h_i(e_{ij}), & D_{ij} = 0, \\ -p_{ij} + h_i(e_{ij}) - m_i(e_{ij}), & D_{ij} = 1; \end{cases} \quad (1)$$

$$\text{Automated assessment: } V_{ij} = -p_{ij} + h_i(e_{ij}) + \beta_i(e_{ij}), \quad D_{ij} = 0.$$

The two human cases are settlement without a bribe and a paid concession. Payment utility is locally linear; the remaining terms are additional valuations beyond cash expenditure.

3.2 Choice and acceptance of a concession

Let $\mathcal{J}_i$ be the taxpayer's feasible alternatives. The chosen alternative maximises the payoff:

$$j_i^* \in \arg \max_{j \in \mathcal{J}_i} V_{ij}.$$

For the same initial state e , denote the inspector alternative without a bribe by H_0 , and the paid concession by H_1 . The net saving from accepting the concession is

$$s_i \equiv p_{iH_0} - p_{iH_1} = T_{iH_0} - (T_{iH_1} + B_{iH_1}).$$

Acceptance is optimal relative to the bribe-free alternative when

$$s_i \geq m_i(e). \quad (2)$$

Thus the acceptance threshold, denoted by $s_i^*(e)$, satisfies

$$s_i^*(e) = m_i(e), \quad m_i(O) < m_i(0) \implies s_i^*(O) < s_i^*(0).$$

Overassessment lowers the required saving precisely when it lowers the additional transaction cost. General dislike of an excessive assessment does not by itself produce this change: $h_i(e)$ is common to the two alternatives and cancels.

3.3 Testable conditions

Conditions C1–C4 distinguish the behavioural explanations. Section 6.2 maps each condition into an empirical restriction, and Section 7.5 reports the corresponding evidence. Reluctance requires a positive transaction cost.

Economic condition	Parameter restriction	Behavioural implication
C1. Savings alone	$m_i(e) = 0$ for all e	Any positive net saving is enough to prefer a paid concession at the same state.
C2. Fixed transaction valuation	$m_i(e) = m_i(0)$	The saving required for acceptance is invariant to the assessment state.
C3. Overassessment reduces transaction reluctance	$m_i(O) < m_i(0)$	A smaller net saving is sufficient following overassessment.
C4. Errors raise the relative value of an inspector	$\beta_i(O) < \beta_i(0)$	Automation is less attractive relative to human assessment after overassessment, at otherwise comparable terms.

At equal payment and a common state without a bribe, automation's payoff advantage is $\beta_i(e)$. A paid inspector alternative must therefore save at least $\beta_i(e) + m_i(e)$. Preference for automation can coexist with acceptance of a cheaper concession. Appendix B derives these payoff comparisons and acceptance thresholds, including the case of different initial assessment states.

If overassessment makes a concession seem more justified, other transaction concerns held fixed, $m_i(O) < m_i(0)$. If errors raise the value of an inspector, other assessor concerns held fixed, $\beta_i(O) < \beta_i(0)$. Neither sign is imposed.

Comparing paid concessions across states alone combines $h_i(e)$ with changes in $m_i(e)$; separating them also requires bribe-free alternatives. Fear, inconvenience, distrust and perceived correction access can all affect these net valuations. The conditions are testable without assigning responses a unique psychological cause.

4 Data and Procedures

4.1 Owners' survey: sampling and recruitment

The owner survey draws on an existing sample of properties with independent remote-sensing measurements. These measurements underpin the offer of independent information about respondents' own properties. Sampling within this frame therefore links the survey experiment, demand for property evidence and the administrative analysis to the same properties, without treating the administrative record itself as the benchmark for assessment accuracy.

The remote-sensing sample comprises 10,564 properties across 48 of Lahore's Excise and Taxation circles, selected from an administrative register of approximately 600,000 properties. Using ESRI satellite imagery from May 2026, the research team independently delineated parcel boundaries, calculated land area from the resulting polygons and predicted covered area using

a machine-learning model. These measurements link to administrative records through Property Identification Numbers. They provide a cross-sectional benchmark for the same identified properties, rather than annual measurements of their physical characteristics.

The measurement frame is spatially broad but selective on property type because dense, contiguous and irregular building layouts constrain parcel delineation (Figure A.2). Table 2 reports the resulting coverage differences. Relative to non-selected properties, the remote-sensing sample contains larger plots, greater covered area and higher liabilities; normalised differences for these characteristics range from 0.20 to 0.32. Comparisons with non-selected organised or gated properties are closer, with normalised differences across the reported characteristics of 0.08–0.14. These comparisons describe observable differences in coverage; the survey findings pertain to participating owners within the imagery-covered frame, rather than all Lahore taxpayers.

Table 2: Remote-sensing sample and comparison properties

	RS selected	Organised/ gated	Non- selected	Normalised difference	
				vs org./ gated	vs non- selected
Land area (sq. yd)	392.7	276.3	179.4	0.14	0.25
Covered area (sq. ft)	3,497	2,723	1,736	0.09	0.20
Tax liability (Rs)	9,670	7,986	4,184	0.09	0.32
Population density	118.6	117.3	114.8	0.08	0.21
Properties	10,564	156,983	546,108		

Notes: Group means from the project comparison. Organised/gated properties are non-selected properties recorded as organised, planned or gated; they are a subset of all non-selected properties. The normalised difference is $(\bar{x}_{RS} - \bar{x}_g) / \sqrt{(s_{RS}^2 + s_g^2)/2}$. Density is reported in the source units. The comparison extract and the historical sampling frame differ in coverage.

Using this sampling frame, for the owner survey, interview targets were specified by tax circle, locality and administrative land-area category. Eligible properties were grouped into four land-area bands: 1–5 marla, above 5 through 10 marla, above 10 through 20 marla, and above 20 through 40 marla. The overall target of 2,100 interviews was allocated across localities in proportion to their aggregate recorded covered area. Within each locality, targets were apportioned across land-area-by-storey cells in proportion to their eligible property counts. Properties were randomly ordered within each cell to construct primary and reserve lists. Out of the total sampling frame of 10,564 properties, fieldwork yielded 2,100 completed interviews from 6,088 approached properties. Table 3 compares completed interviews, approached properties without a completed interview, and properties not approached on observed administrative characteristics. It shows that citizens who completed the surveys are statistically similar to citizens who refused or were not surveyed.

Table 3: Survey participation and observed property characteristics

Variable	(1) Completed Surveys		(2) Refused interview		(3) Not approached		Mean differences					
							(1)–(2)		(1)–(3)		(2)–(3)	
	N	Mean (SD)	N	Mean (SD)	N	Mean (SD)	N	Mean difference	N	Mean difference	N	Mean difference
Land Area (marla)	2100	15.361 (16.588)	3988	16.005 (22.870)	4476	15.604 (35.117)	6088	-0.644	6576	-0.243	8464	0.401
Number of floors	2100	2.870 (0.404)	3988	2.860 (0.394)	4476	2.535 (0.680)	6088	0.010	6576	0.336	8464	0.325
Total covered area (sqft.)	2100	3072.161 (1956.331)	3988	2758.386 (1951.123)	4476	2924.857 (17482.277)	6088	313.776	6576	147.304	8464	-166.472
Total property tax	2100	10118.323 (12897.526)	3988	10016.087 (11196.390)	4476	9144.635 (31565.469)	6088	102.235	6576	973.688	8464	871.453

4.2 Survey Measures and Sequence

Fieldwork was conducted between July and August 2026. The consent protocol explained voluntary participation, confidentiality and the hypothetical nature of the choices; participants were offered Rs 1,000 in mobile credit. Appendix C gives survey questions that were asked of the respondents. The survey measures owners’ experience with the Excise and Taxation Department, familiarity with their property record, perceptions of recorded land and covered area, and trust in officers’ assessments. It distinguishes general beliefs about informal payments from respondents’ own reported experiences. Respondents also report recent assessment notices, whether they regard the assessed amount as accurate, awareness of available responses and intentions to seek checking or correction. These measures describe experience, beliefs and intended behaviour; they do not verify that an assessment is legally incorrect or that an application for correction has been submitted.

Documentary resources are measured by whether an owner can produce, within approximately one week, a registry or sale deed, architectural drawings recording land and covered area, and the latest property-tax receipt or PT-1. These categories describe reported arrangements rather than independently established eligibility to appeal. Before the assessment-choice tasks, respondents report how easy it would be to correct an erroneous computer assessment.

4.2.1 Bayesian Adaptive Assessment-Choice Experiment

The survey embedded a choice experiment. Owners chose between assessment receipts displaying the assessor, a reference correct bill, the initial bill, any informal payment, the final official bill and total expenditure. Inspector offers included paid concessions while satellite offers did not. Choices neither alter tax records nor implement payments. Appendix Figure D.1 reproduces an example receipt pair; Table D.1 reports the four common comparisons and their eligible samples.

The design varied assessor type, assessment error and concession terms, with expenditure equal to the final official bill plus the informal payment. Four common comparisons precede adaptive questions: assessor choice at equal correct bills; acceptance of a concession between two inspectors; and two comparisons of paid inspector offers differing in initial assessment state. The first places the satellite in option A. The latter comparisons impose equal displayed expenditure, with the fourth also imposing equal informal payments.

The choice experiment updated preference distributions in real-time using a Bayesian utility function that used priors from previous surveys and selected subsequent questions for expected information gain. Its selection model separated assessor valuation, paid-transaction cost and the change in that cost under overassessment. The broader estimation model additionally permitted direct assessment-state and position effects. The analysis retains 21,001 valid paired tasks from 2,100 owners. Appendix E sets out the likelihood and assignment assumptions; estimates are conditional on the sequential design rather than independent factorial randomisation.

Independent evidence offer. A separate offer concerns computer-estimated details of the owner’s property at assigned prices of Rs 0, Rs 100 or Rs 200, deductible from the participation payment. The 2,099 substantive responses provide a behavioural measure for the demand for automated assessment. Appendix C gives the question wording and coding. The offer response does not verify delivery.

4.3 Administrative Records and Correction Data

I combine three separate administrative datasets using Property Identification Numbers (PINs): an annual cadastral panel, formal appeal records and PT-1 correction records. These sources provide distinct information on recorded property characteristics, taxpayers’ use of the appeal process and corrections documented by the administration.

Annual cadastral records. The unbalanced panel contains 23,233 residential properties observed between FY2022–23 and FY2026–27. These properties were drawn from the same administrative cadastre of approximately 600,000 residential properties used to construct the remote-sensing sample. The draw follows the same within-circle stratification by recorded covered area, extended to all 194 tax circles in Lahore, and includes the 10,564 properties forming the owner-survey sampling frame. Annual records report land and covered area, with payable-tax fields available through FY2025–26. The panel therefore allows me to follow changes in recorded property characteristics, although not every property has usable observations in every year.

Formal appeal records. The appeals dataset covers 2024–2026 and records whether an appeal was made, its filing date, its recorded status and the office handling the case. These fields identify recorded use of the formal appeal route and the administrative status of the resulting

cases. Filing an appeal is distinguished from obtaining a correction; the presence of an appeal record alone does not establish that the assessment was amended.

PT-1 correction records. A separate dataset contains corrections documented in PT-1 records during 2024–2026. These records contain property characteristics together with comments from the Excise and Taxation Officer (ETO) describing the correction. The comments provide information about the nature of the recorded change that is not available from comparing annual cadastral entries alone. Linking these records to the appeals dataset identifies properties with both a documented correction and a recorded appeal, while retaining the distinction between the two sources. A property-level match does not, by itself, establish that a particular correction resulted from that appeal.

The linked sources distinguish recorded appeals, documented PT-1 corrections and changes in annual property attributes.

5 Outcome measures

Assessor choice on equal terms. The first common task records whether an owner chooses satellite rather than inspector assessment when both initial bills are correct, total expenditure is equal and neither offer includes an informal payment. The outcome is one for satellite choice and zero for inspector choice; all 2,100 owners are eligible. It describes the offers as presented, with the satellite in option A (See Figure D.2).

Acceptance of a paid concession. The second common-task outcome is one for choosing a paid inspector offer over an unpaid inspector offer, and zero otherwise. Both initial bills must be correct. These displayed conditions hold for 2,050 owners. Net savings equal the official bill reduction less the informal payment. I also report this outcome separately by the owner's first-task choice (See Figure D.3).

Choice of an initially excessive bill at equal expenditure. The third outcome is one for choosing the paid inspector offer with an initially excessive bill rather than the paid inspector offer with a correct initial bill. Eligibility requires equal displayed total expenditure, yielding 927 owners. This comparison changes the assessment state while holding the eventual total payment fixed (See Figure D.4).

Choice of an initially excessive bill at equal expenditure and payment. The fourth outcome applies the preceding definition with the additional requirement of equal informal payments, yielding 1,624 owners. These are separate common-task comparisons, with separate denominators (See Figure D.5) (Appendix Table D.1).

Choices across the full sequence. For owner i , alternative $j \in \{A, B\}$ and task t , y_{ijt} is one for the selected receipt and zero otherwise. The analysis contains 21,001 paired tasks. Let R_i be the displayed correct reference bill, $\tilde{T}_{ijt}$ the initial bill, T_{ijt} the final official bill and B_{ijt} the informal payment. Receipt attributes are

$$O_{ijt} = \mathbf{1}\{\tilde{T}_{ijt} > R_i\}, \quad U_{ijt} = \mathbf{1}\{\tilde{T}_{ijt} < R_i\}, \quad D_{ijt} = \mathbf{1}\{B_{ijt} > 0\}, \quad P_{ijt} = (T_{ijt} + B_{ijt})/1000.$$

S_{ijt} indicates satellite assessment. Thus O and U describe the initial assessment, D identifies a paid concession, and P measures expenditure in thousands of rupees. A bill reduction without an informal payment is not a paid concession.

Acceptance of independent evidence (S6.1). The outcome equals one for accepting computer-estimated details of the owner's property and zero for declining, at the recorded offer of Rs 0, Rs 100 or Rs 200. There are 2,099 substantive responses. Acceptance does not establish delivery, use or successful correction.

6 Empirical strategy

6.1 Taxpayers' readiness to seek correction: descriptive estimation

I begin with descriptive evidence on correction intentions, reasons for low intention, digital comfort and access to documents. These survey measures describe the setting in which owners choose assessors and respond to the evidence offer. I summarise each measure separately, retaining its own eligible sample. Readiness provides an organising theme for these comparisons; I do not construct a combined readiness score.

For survey measure m and response category or stated grouping r , I estimate

$$\hat{p}_{mr} = 100 \frac{n_{mr}}{n_m}, \quad n_{mr} = \sum_{i \in S_m} \mathbf{1}\{Q_{im} \in r\}, \quad (3)$$

where Q_{im} is owner i 's response, S_m contains eligible owners with substantive answers, and $n_m = |S_m|$. Thus $\hat{p}_{mr}$ is the percentage reporting category r within the relevant valid-answer sample. The reported comparisons are unweighted sample proportions.

For intention (S9.1), eligibility requires a recent assessment communication (S3.6) and an assessment regarded as somewhat or much too high (S3.7). I report the shares scoring 0–4 and 5–10 on the 0–10 intention scale, described as low and high intention, respectively. For reasons (S9.2), I further restrict the sample to those reporting low intention and estimate a separate share for each stated main reason: perceived fairness, cost, lack of procedural knowledge, fear, hassle or another reason.

Digital comfort (S1.6) and document difficulty (S3.17) use all completed interviews with a substantive answer to the respective question. Low digital comfort comprises “not at all comfortable” and “a little comfortable”; document difficulty comprises “very difficult”, “difficult” and “somewhat difficult”. Appendix C provides the wording and coding. Tables C.2–C.5 retain all original response categories and separately show unknown, refused and missing answers. Their percentages use the full eligible sample, including nonresponse; the summary percentages in Section 7.1 use the substantive-answer denominators in equation (3).

These estimates describe reported intentions and circumstances among the surveyed owners. The intention and reason shares concern the specified subgroups; digital comfort and document difficulty concern the broader survey sample. They do not establish actual applications, successful correction or causal effects on choice.

6.2 BACE and the model’s testable conditions

I use the repeated choices to examine conditions C1–C4 in Section 3.3. The specification separates general responses to an incorrect initial bill from changes in the valuation of the assessor or a paid transaction:

$$v_{ijt} = \alpha S_{ijt} + b_p P_{ijt} + \kappa D_{ijt} + \gamma_O O_{ijt} + \gamma_U U_{ijt} \\ + \delta_O S_{ijt} O_{ijt} + \delta_U S_{ijt} U_{ijt} + \lambda_O D_{ijt} O_{ijt} + \lambda_U D_{ijt} U_{ijt} + \pi \mathbf{1}\{j = B\}. \quad (4)$$

An unpaid inspector with a correct initial bill is the reference. At common expenditure, α compares satellite with inspector assessment and κ compares paid with unpaid inspector offers at a correct bill. The γ_e coefficients describe initial errors under unpaid inspector assessment; δ_e and λ_e describe how assessor and transaction valuations change with that state. The position term allows option B to differ from option A.

With type-I extreme-value choice shocks, the probability of choosing B is

$$\Pr(y_{iBt} = 1 \mid X_{it}) = \frac{\exp(v_{iBt} - v_{iAt})}{1 + \exp(v_{iBt} - v_{iAt})}. \quad (5)$$

I report a pooled logit with owner-clustered standard errors, a panel mixed logit with persistent owner-specific satellite and paid-deal coefficients, and a pooled specification restricted to tasks five onwards. The mixed model integrates over each owner’s complete sequence and retains a common expenditure coefficient. Its transaction tests concern mean coefficients, not the stronger restriction that transaction valuation is zero for every owner.

For a positive utility scale μ , the mapping to the model is $b_p = -1000\mu$, $\alpha = \mu\beta(0)$, $\kappa = -\mu m(0)$, $\gamma_e = \mu h(e)$, $\delta_e = \mu[\beta(e) - \beta(0)]$ and $\lambda_e = -\mu[m(e) - m(0)]$. This gives four empirical counterparts:

1. **C1: savings alone.** $m(e) = 0$ in every state requires $\kappa = \lambda_O = \lambda_U = 0$. I test these restrictions jointly; in the mixed model they apply to mean transaction valuation.

2. **C2: fixed transaction valuation.** $m(e) = m(0)$ requires $\lambda_O = \lambda_U = 0$. The joint test asks whether the saving required for acceptance changes with the initial assessment state.
3. **C3: overassessment lowers the required saving.** $m(O) < m(0)$ implies $\lambda_O > 0$. I assess the estimated sign and its 95% interval. Comparing λ_O with λ_U additionally asks whether this shift differs from the response to underassessment.
4. **C4: overassessment raises the relative value of an inspector.** $\beta(O) < \beta(0)$ implies $\delta_O < 0$. I assess this sign and interval separately from the broader joint restriction $\delta_O = \delta_U = 0$.

Table 5 reports the joint tests and directional estimates in this order. Intervals use two-sided normal inference, so a failure to exclude zero does not establish equality. Direct error responses are γ_e for unpaid inspectors, $\gamma_e + \lambda_e$ for paid inspectors and $\gamma_e + \delta_e$ for satellites. These contrasts use the full covariance matrix and retain $h(e)$; an aversion to incorrect bills alone does not establish C3 or C4.

Adaptive offers depend on earlier choices. Interpretation requires assignment to depend on recorded history without additional dependence on current unobserved choice shocks, together with the maintained preference specification. The later-task specification checks sensitivity to the common opening comparisons. Appendix E gives the likelihood and additional specifications.

6.3 Demand for independent evidence

For 2,099 substantive S6.1 responses, I estimate

$$100\text{Buy}_i = \alpha_B + \tau_{100}\mathbf{1}\{q_i = 100\} + \tau_{200}\mathbf{1}\{q_i = 200\} + \varepsilon_i. \quad (6)$$

Each τ compares acceptance at a paid offer with free provision, in percentage points; HC3 standard errors allow different variances across groups. This saturated specification imposes no linear price slope. I interpret the recorded price comparisons descriptively because the available implementation documentation does not fully verify deployed assignment. They measure acceptance of evidence, not its subsequent use.

6.4 Assessor choice and expected correctability

The outcome is the BACE choice indicator y_{iBt} , equal to one when owner i selects option B in task t and zero when the owner selects A. Expected correctability (S5.5) enters as an explanatory

variable. I estimate the pooled choice logit

$$\log\left(\frac{p_{it}}{1 - p_{it}}\right) = \sum_{r=1}^3 \mathbf{1}\{G_i = r\} \Delta X'_{it} \boldsymbol{\theta}_r + \Delta S_{it}(\rho_H H_i + \rho_E E_i + \rho_L L_i), \quad (7)$$

where $p_{it} = \Pr(y_{iBt} = 1 \mid X_{it}, H_i, E_i, L_i, G_i)$, $\Delta X_{it} = X_{iBt} - X_{iAt}$ contains the differences in the ten baseline choice terms in equation (4), and $\Delta S_{it} = S_{iBt} - S_{iAt}$. From S5.5, H_i indicates that correcting an erroneous computer assessment is expected to be hard and E_i that it is expected to be easy or very easy; very hard is the reference category. L_i indicates little or no officer trust in S3.9, and G_i denotes the recorded information group. Appendix C gives both questions and their coding.

Appendix Table E.5 reports three specifications: common baseline coefficients without trust (2,032 owners; 20,320 tasks); addition of trust (2,004 owners; 20,040 tasks); and equation (7), allowing all ten baseline coefficients to vary across the three information groups on the same 2,004-owner sample. Standard errors are clustered by owner. Positive ρ_H or ρ_E indicates greater relative valuation of satellite assessment than among owners expecting correction to be very hard; these are logit coefficients, not percentage-point effects. Owner-level beliefs enter through satellite interactions because terms common to both alternatives cancel from the utility difference. Correctability is elicited after information scripts and is not randomly assigned. The coefficients describe an association with a possible component of $\beta_i(0)$, rather than the effect of providing recourse or C4's response to a displayed error.

For Figure 6, I separately define $Y_i^{\text{sat},1} = 1$ when the owner chooses the satellite in the first task and zero when the owner chooses the inspector. Both options display equal, correct bills. I report the mean of this outcome within each S5.5 category, with Wilson 95% intervals.

7 Results

7.1 Correction in practice and taxpayer readiness

I begin with the correction setting. Administrative remarks document correction and verification within the assessment process. Survey responses describe owners' intention to seek correction, reasons for low intention, digital comfort and difficulty locating property documents. These findings provide context for the subsequent analysis of assessor choices and demand for independent evidence.

Recorded correction and verification. Among 33 property records selected for area correction or verification wording, none has a structured appeal indicator, two mention appeal in the re-

marks, and 31 have neither form of appeal evidence.¹ Some remarks describe document checks or site verification. Structured appeal fields alone therefore miss documented activity within the assessment process.

These are selected records, not 33 verified new corrections. One historical order changes a unit from 75 marla to 75 square yards although the property is already recorded at 3 marla throughout the panel. Other entries cannot be checked against complete measurements. Missing appeal evidence does not prove that no appeal occurred. The finding establishes a documented administrative function; it does not assign every entry to a new event. In terms of the model, the net value of an assessor can include help with correction alongside any private concession, although these records do not measure the size of that component in $\beta_i(e)$.

Intention to seek checking or correction. Among 169 owners who received a recent assessment communication and considered it too high, the average stated likelihood of seeking checking or correction is 4.95 out of ten (SD 1.02; 95% CI 4.79–5.10), with a median of five. All 169 provide substantive S9.1 answers. Forty-five owners (26.6%) score 0–4 and 124 (73.4%) score 5–10. Figure 1, Panel A, displays the low-intention share. These responses measure prospective intention, not an application or its outcome.

Reasons for low intention. Among the 45 low-intention owners, 27 (60.0%) cite hassle, 16 (35.6%) cite fear of consequences and two (4.4%) say the assessment seems fair (S9.2). None selects cost, lack of procedural knowledge or another reason. All 45 provide substantive answers. Panel B reports the separate shares and their uncertainty; these are stated reasons, not verified barriers to submitted applications.

Digital comfort. Of 2,081 substantive S1.6 respondents, 754 (36.2%) report little or no comfort with a simple online form. Nineteen of the 2,100 completed interviews supply a “don’t know” answer and are excluded from this share. This measures reported comfort rather than tested digital ability.

Difficulty locating documents. Of 2,070 substantive S3.17 respondents, 1,343 (64.9%) report at least some difficulty locating property documents. The remaining 30 completed interviews contain one “don’t know” response and 29 refusals. Panel A displays this share alongside low digital comfort, retaining each measure’s denominator. Reported difficulty does not establish legal exclusion from a correction procedure.

¹The screen requires both an area term and correction or verification wording. Area terms include land area, covered area, common spellings of square feet and square yards, and marla. Correction terms include correction, mistake, error, rectification, wrong, excess, overassessment, overstatement and verification. Appeal wording allows spelling variations and appeal-related references. Generic Section 9 or Assessing Authority references alone are insufficient because they may concern ownership changes, subdivisions or other actions.

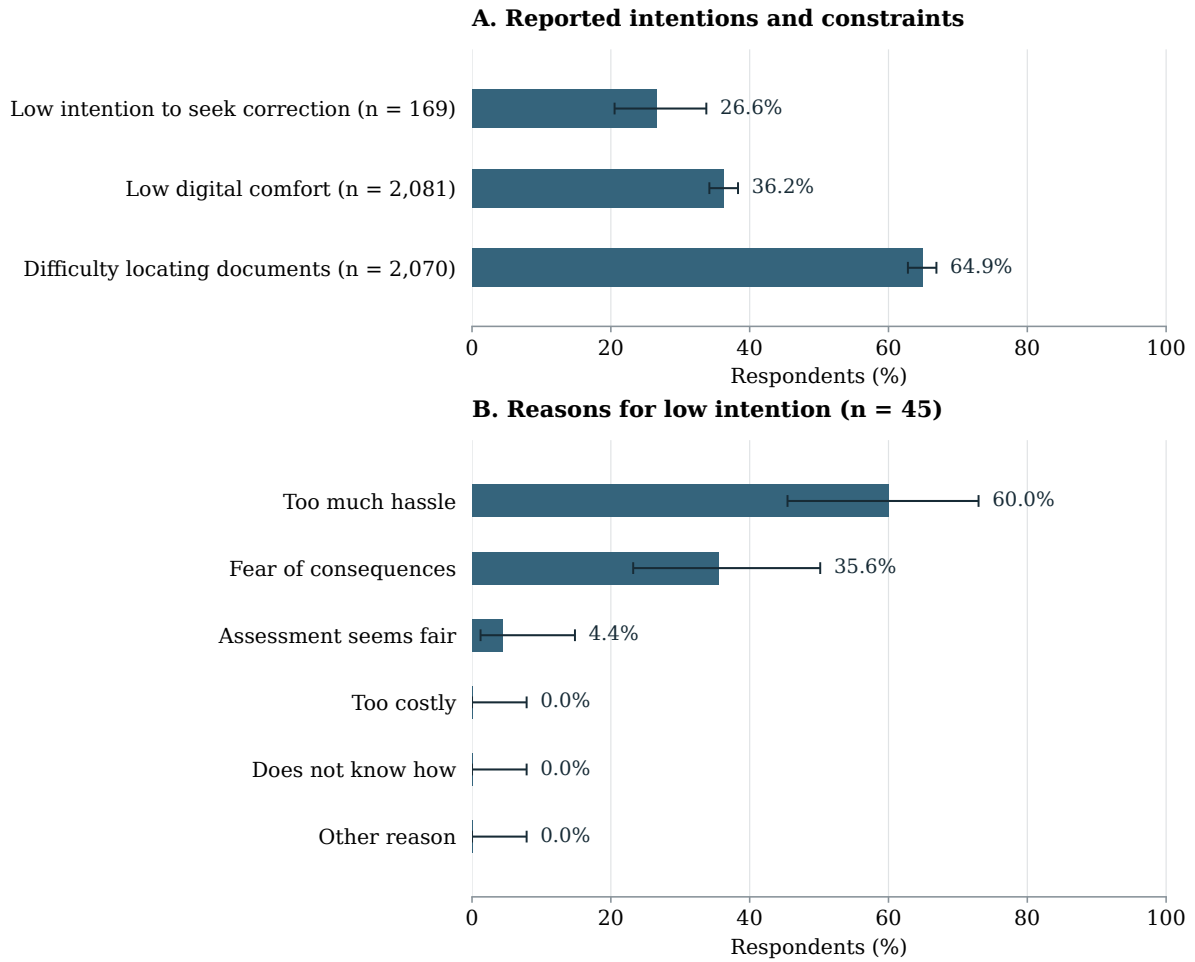


Figure 1: Reported readiness to seek correction

Notes: Bars report unweighted shares; whiskers are pointwise Wilson 95% confidence intervals. Panel A: low intention is an S9.1 score of 0–4 among 169 recent assessment recipients judging the assessment too high; low digital comfort is little or no comfort in S1.6 among 2,081 substantive respondents; document difficulty includes any difficulty in S3.17 among 2,070 substantive respondents. Panel B reports each S9.2 main reason among the 45 low-intention owners; all provide substantive answers. The samples differ across measures. Zero observed selections do not establish zero prevalence.

7.2 Assessor preference on equal terms

At equal correct bills, 1,256 of 2,100 owners choose satellite assessment: 59.8%, with a 95% interval of 57.7–61.9% (Figure 2). This establishes a majority for the automated offer as presented. The satellite always appears in option A in this task. After accounting for position across the full sequence, its coefficient at a correct bill is positive but imprecise (Table 4). The common-task share therefore does not establish a position-independent preference for every form of automation.

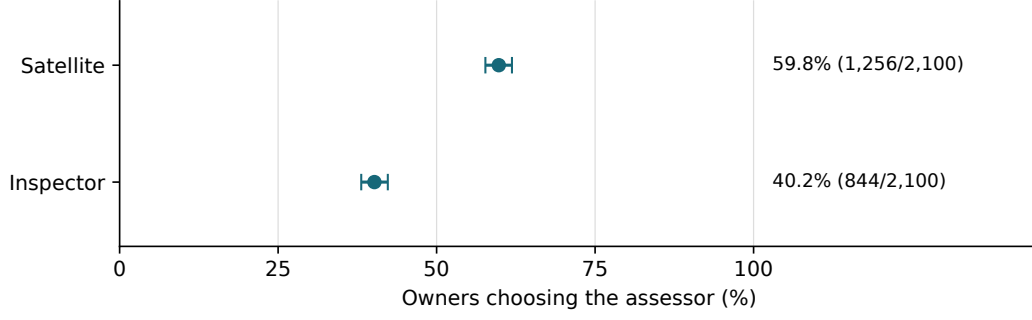


Figure 2: Assessor choice at equal correct bills

Notes: Both offers have the same correct initial bill, equal total expenditure and no informal payment. Counts are out of 2,100 owners; bars show Wilson 95% confidence intervals. The satellite appears in option A.

7.3 Acceptance of a paid concession

When the next eligible comparison offers a paid inspector concession, 1,122 of 2,050 owners accept it (54.7%). Acceptance is 58.5% among the 1,218 eligible owners who initially chose the satellite and 49.3% among the 832 who initially chose the inspector (Figure 3). Thus preference for satellite assessment on equal terms coexists with acceptance of an inspector's offer when its financial terms change. The median displayed net saving is Rs 1,200, with a range from zero to Rs 33,100.

Higher total expenditure reduces choice in all three specifications (Table 4). The model explains why these choices can coexist: the first comparison concerns $\beta_i(0)$, while the second concerns $s_i - m_i(0)$ (equation (2)). A taxpayer can value automation on equal terms and still accept a concession over an unpaid inspector offer. The second task does not compare that concession directly with a satellite offer. Section 7.5 tests whether savings alone account for the choices.

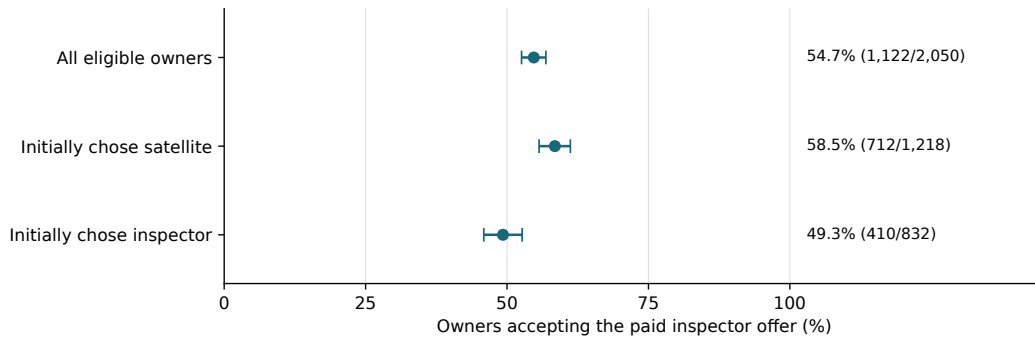


Figure 3: Acceptance of a paid inspector concession

Notes: The outcome is selection of the paid rather than unpaid inspector offer at correct initial bills. All eligible owners: 1,122/2,050; initial satellite choosers: 712/1,218; initial inspector choosers: 410/832. Subgroups condition on the first choice. Intervals are Wilson 95%.

Table 4: Choices across all valid assessment offers

Attribute	Pooled logit	Panel mixed logit	Later tasks
Satellite, correct bill	0.056 (0.047)	0.074 (0.063)	0.221 (0.136)
Total expenditure, Rs 1,000	-0.097*** (0.017)	-0.287*** (0.032)	-0.115*** (0.022)
Overassessment, unpaid human	-0.625*** (0.089)	-0.971*** (0.109)	-0.245 (0.153)
Underassessment, unpaid human	-0.588*** (0.087)	-0.489*** (0.102)	-0.359** (0.148)
Satellite $\times$ overassessment	0.093 (0.088)	-0.114 (0.109)	-0.065 (0.159)
Satellite $\times$ underassessment	0.142* (0.082)	0.112 (0.098)	0.001 (0.154)
Paid deal, correct bill	0.381*** (0.046)	0.142** (0.063)	0.316** (0.140)
Deal $\times$ overassessment	0.050 (0.084)	0.289*** (0.103)	0.146 (0.160)
Deal $\times$ underassessment	0.007 (0.083)	0.281*** (0.099)	0.073 (0.158)
Option B	-0.244*** (0.019)	-0.302*** (0.023)	-0.126*** (0.021)
Tasks	21,001	21,001	12,601
Respondents	2,100	2,100	2,100

Notes: Coefficients are in logit units; standard errors in parentheses use respondent clustering or respondent likelihood scores. The mixed model integrates persistent normally distributed satellite and deal coefficients over each owner's complete sequence. Later tasks are slots 5 onward. All specifications use displayed expenditure in thousands of rupees and include the terms shown. No coefficient is labelled a pure moral cost or bargaining parameter. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$; two-sided normal-reference Wald tests using the reported standard errors. Stars refer to individual coefficients, not joint tests.

7.4 Preferences over assessment states

Initially incorrect bills are less attractive even after conditioning on eventual expenditure. In the pooled model, overassessment and underassessment reduce unpaid-inspector utility by 0.625 and 0.588 logit units. Both error contrasts are also negative for paid inspectors and satellites (Figure 4). The later-task overassessment contrast for paid inspectors is less precise and includes zero. The figure presents comparisons within each model; coefficient magnitudes across logit specifications need not have a common scale.

The common comparisons provide a direct counterpart: 219 of 927 owners (23.6%) choose the initially overassessed paid offer at equal expenditure, and 423 of 1,624 (26.0%) do so when the informal payment is also equal (Appendix Table D.1). The model writes this comparison as $h_i(O) - [m_i(O) - m_i(0)]$ (Appendix B). A low share can therefore reflect dislike of the initial error even if overassessment reduces the transaction cost. The comparison alone cannot establish whether that acceptance threshold changes.

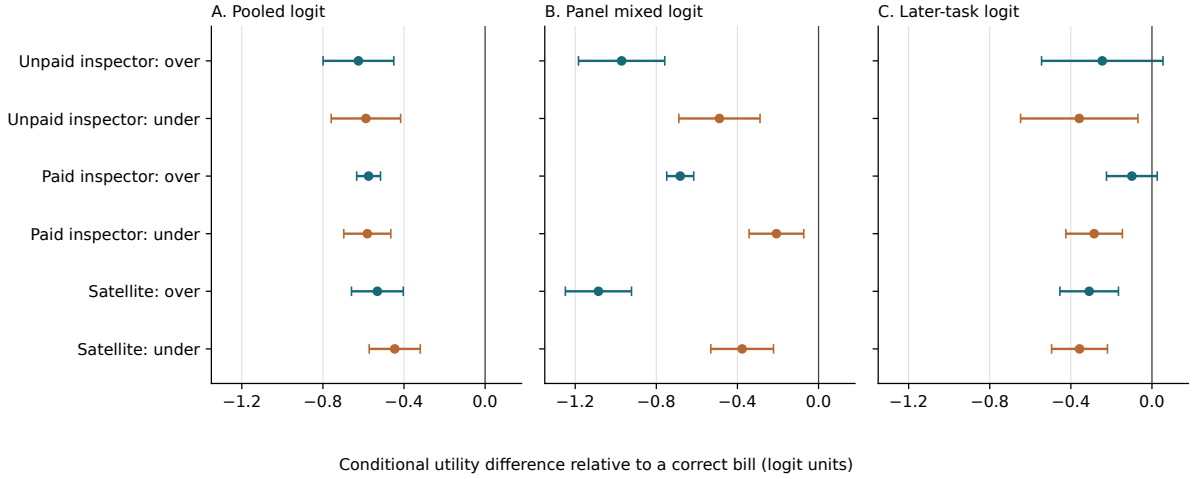


Figure 4: Conditional valuation of initially incorrect assessments

Notes: Each point compares an initial error with a correct bill for the same assessment arrangement, conditional on total expenditure and position. Intervals use the full coefficient covariance matrix. Separate panels report pooled, panel mixed and later-task models. The zero line denotes no conditional difference; coefficients are in model-specific logit units.

7.5 Evidence on the model's testable conditions

Table 5 returns to conditions C1–C4 in Section 3.3, using the coefficient mapping in Section 6.2.

C1: savings alone is rejected. The restriction $\kappa = \lambda_O = \lambda_U = 0$ is rejected in all three specifications ($p < 0.001$). Conditional on total expenditure and state, the paid-transaction terms contribute to choice. At a correct bill, κ is positive, mapping to a negative $m(0)$ in the homogeneous model and a negative mean in the mixed model. Thus the evidence does not support interpreting the transaction term uniformly as reluctance. Its sign does not establish an intrinsic preference for bribery, and the mixed-model test concerns means rather than every owner's preferences.

C2: evidence against fixed transaction valuation depends on specification. The restriction $\lambda_O = \lambda_U = 0$ is rejected in the mixed model ($p = 0.0011$), but not in the pooled or later-task models ($p = 0.836$ and $p = 0.615$). The evidence that the acceptance threshold $m(e)$ changes with the assessment state therefore depends on allowing persistent preference heterogeneity.

C3: a lower threshold after overassessment has limited support. The mixed-model estimate of λ_O is 0.289 (95% interval: 0.088–0.491), consistent with $m(O) < m(0)$ and a smaller saving required to accept a paid concession. The pooled and later-task estimates are also positive, but their intervals include zero. Moreover, $\lambda_O = \lambda_U$ is not rejected in any specification. The mixed-model finding therefore does not distinguish a response specific to overassessment from a similar response to underassessment, or identify a moral-justification mechanism.

C4: an increased relative value of inspectors is not established. The predicted sign is $\delta_O < 0$. Its estimate is positive in the pooled model and negative in the mixed and later-task models, with all three intervals including zero. The broader state-invariant-assessor restriction is also not rejected. The data therefore do not establish that overassessment reduces the satellite’s relative valuation $\beta(O)$, although they do not demonstrate invariance either. The negative error contrasts in Figure 4 concern valuation of the assessment state, which the model distinguishes from this change in relative assessor value.

Together, the tests reject the savings-alone account more consistently than they support state-dependent transaction or assessor valuations. These conclusions concern the model’s net valuations, not a unique psychological explanation. Appendix Table E.4 further shows that payment composition matters conditional on total expenditure, qualifying the compact model’s exclusion of an additional bribe-amount term.

Table 5: Evidence on the model’s testable conditions

Condition or restriction	Pooled	Mixed	Later tasks
<i>Panel A. Conditions C1–C2: joint-test p-values</i>			
C1. Savings alone: $\kappa = \lambda_O = \lambda_U = 0$	< 0.001	< 0.001	< 0.001
C2. Fixed transaction valuation: $\lambda_O = \lambda_U = 0$	0.8361	0.0011	0.6154
<i>Panel B. Conditions C3–C4: coefficients and 95% intervals</i>			
C3. Lower saving threshold: $\lambda_O > 0$	0.050 [-0.114, 0.214]	0.289 [0.088, 0.491]	0.146 [-0.169, 0.460]
C4. Greater inspector value: $\delta_O < 0$	0.093 [-0.079, 0.266]	-0.114 [-0.327, 0.099]	-0.065 [-0.376, 0.246]
<i>Panel C. Supplementary restrictions: p-values</i>			
State-invariant assessor valuation: $\delta_O = \delta_U = 0$	0.1684	0.2287	0.8161
Equal transaction shifts: $\lambda_O = \lambda_U$	0.6865	0.9506	0.4986

Notes: Conditions correspond to Section 3.3; their empirical mapping is in Section 6.2. Panels A and C report saved Wald tests using full covariance matrices, with 3, 2, 2 and 1 degrees of freedom, respectively. Panel B reports existing coefficient estimates and two-sided normal-reference intervals; the inequalities state the model predictions, not one-sided p -values. The mixed model tests mean transaction valuation; savings alone for every owner would additionally require zero paid-deal heterogeneity. Inference uses respondent clustering or respondent likelihood scores and is unadjusted for multiple comparisons. Nonrejection does not establish equality.

7.6 Demand for independent property evidence

Acceptance is 94.8% when evidence is free, 39.0% at Rs 100 and 39.4% at Rs 200 (Figure 5). The paid-price differences from free provision are –55.7 and –55.4 percentage points, each with an HC3 standard error of approximately 2.0 points (Appendix Table F.2). Demand falls sharply between free and paid provision, with little further difference between the two positive

offers. Acceptance at a positive price demonstrates willingness to obtain the offered information, without establishing that it is delivered or used to secure correction.

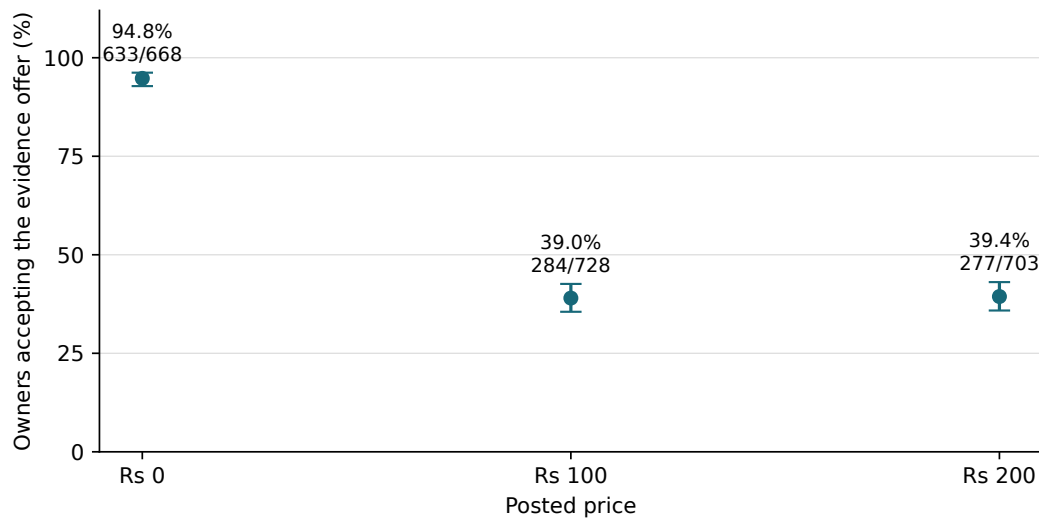


Figure 5: Acceptance of independent property evidence by offer price

Notes: Question S6.1. Acceptance is 633/668 at Rs 0, 284/728 at Rs 100 and 277/703 at Rs 200. One additional owner at Rs 200 supplies no substantive answer. Intervals are Wilson 95%. The outcome is acceptance of the offer, not verified delivery or use.

7.7 Expected correctability and assessor choice

Satellite choice at equal bills rises from 41.6% among owners expecting correction to be very hard, to 62.0% among those expecting it to be hard, and 89.6% among those expecting it to be easy or very easy (Figure 6). Positive satellite interactions persist across the full sequence after adjusting for officer trust and allowing choice coefficients to differ by information group (Appendix Table E.5). In the model, anticipated recourse can contribute to $\beta_i(0)$ even when the displayed bill is correct. This association concerns differences between owners in expected correctability; C4 instead concerns how relative assessor valuation changes with the displayed assessment state. The former does not establish the latter. Correctability can also capture broader confidence in the system, so the association does not identify the effect of providing a correction channel.

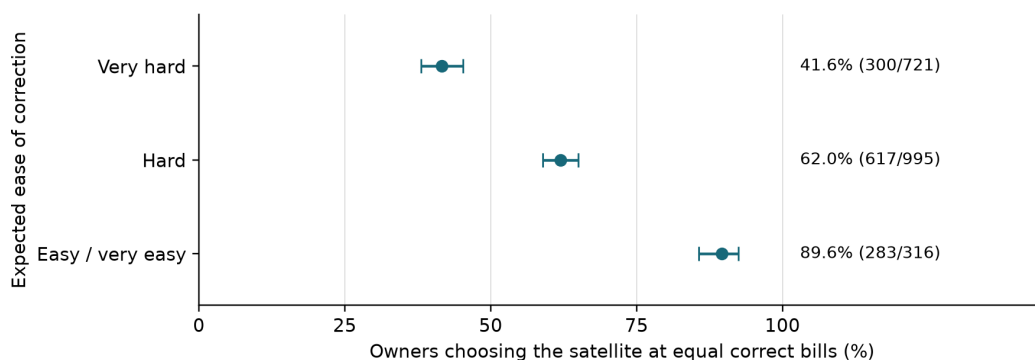


Figure 6: Satellite choice by expected correctability

Notes: Question S5.5 classifies expected ease of correcting a computer error. Satellite-choice counts are 300/721 (very hard), 617/995 (hard) and 283/316 (easy or very easy). Intervals are Wilson 95%. This is an observational comparison among 2,032 substantive respondents.

8 Conclusion

Technological reform changes how citizens encounter the tax system. Under personal administration, the same official may explain a liability, establish the facts about a property, correct a record and offer a private concession. Taxpayers' responses to this relationship cannot be inferred from a general preference for automation. The choice results distinguish preferences over assessors from the terms of particular transactions, while the administrative and survey evidence highlights access to correction. Together, they suggest that revenue authorities pursuing less personal taxation must address how citizens understand, challenge and resolve assessments, not only how the state calculates them.

This distinction is especially consequential where trust in officials is low, as documented in this study's baseline surveys. Dissatisfaction with officials does not automatically legitimise a computer-generated bill. Taxpayers may welcome fewer opportunities for arbitrary bargaining yet remain uncertain about who will address incorrect measurements or unexplained changes in liability. The association between expected correctability and satellite choice is consistent with this concern, although it does not establish that providing a correction channel causes acceptance. It cautions against treating reduced personal contact as sufficient to earn confidence. The challenge for revenue authorities is to make procedures dependable even where trust in individual officials is weak.

The model explains why a preference for an impersonal assessor can coexist with acceptance of a financially attractive concession: the decisions compare different alternatives. The tests reject a savings-alone account of conditional choices but provide less consistent support for the claim that overassessment specifically lowers transaction reluctance or increases the relative value of inspectors. These findings do not justify retaining discretionary negotiation as a general remedy for errors. Instead, they motivate separating the legitimate functions of personal administration from private arrangements to alter liabilities. Explanation, verification and non-appeal

administrative correction need clearly assigned responsibilities and reviewable procedures as bargaining is reduced.

This challenge extends beyond property taxation as the range of available technologies expands. The OECD documents digital identity, third-party data exchange, electronic invoicing, automated responses and artificial intelligence, although its comparative report primarily covers members of the Forum on Tax Administration (OECD, 2025). These technologies can change several administrative functions at once, making sequencing consequential. An authority's capacity to detect discrepancies and issue assessments may expand before taxpayers have equally usable means to check the underlying information or obtain a response. This imbalance is a risk of transition design, not an inevitable consequence of technology.

A practical starting point is to make the basis of an assessment visible to the person expected to pay it. In property taxation, a notice should identify the property, the characteristics and valuation inputs used, the applicable calculation, and the reason for any change from the previous bill. Taxpayers should be able to inspect the relevant record and obtain an understandable explanation of measurement discrepancies. Where imagery or another external source contributes to assessment, its date and relevant measurements should be available in a form that can be questioned. The observed demand for independent property evidence underscores the importance of access to this information. It does not establish that such evidence is legally decisive or will secure correction. A reproducible calculation must therefore be accompanied by a way to contest its inputs.

Human involvement remains an institutional design choice even as calculation becomes more automated. Local knowledge and judgement may be needed to distinguish record errors from genuine changes in taxable characteristics or assess conflicting evidence. Authorities could assign these tasks to identifiable staff while recording the evidence considered, reasons for amendments and any supervisory approval. Reviewing unusual amendment patterns alongside unresolved complaints would support oversight without treating every reduction in liability as misconduct. As new data and analytical tools are introduced, access permissions, records of system changes and responsibility for correcting faulty inputs should also be explicit. The OECD's discussion of data governance and oversight of artificial intelligence reinforces the importance of these capabilities (OECD, 2025).

For resource-constrained authorities, a feasible approach is to introduce assessment technology and correction arrangements together in selected locations, learn from their operation and expand in stages. Evaluation should follow taxpayers from receipt of an assessment through any request for explanation, verification or review. Alongside revenue and processing speed, useful measures include whether notices are understood, the time and money spent obtaining a response, how requests are resolved, and whether input errors recur. These measures should be examined by documentary resources, digital comfort and relevant taxpayer characteristics. More recorded requests may reflect easier access to correction, while few requests may conceal substantial barriers; neither count alone establishes success. Resolved cases should inform revisions to the records and procedures that generated disputes.

This agenda responds to concerns identified by the study; its effects remain to be established. The experiment measures choices over specified offers, while the administrative and belief analyses describe associations within the observed samples. They do not estimate the revenue, trust or welfare effects of removing inspectors. Future implementation could compare alternative ways of providing assessment explanations, independent evidence and assistance, measuring both correction outcomes and compliance. Punjab's implemented valuation reform should also be distinguished from the prospective reduction in officer involvement examined here. The broader lesson for revenue authorities, particularly in low- and middle-income countries, is that the capacity to respond to taxpayers should develop alongside the capacity to assess them. More impersonal taxation becomes a credible institutional improvement when decisions can be explained, errors corrected at reasonable cost, and responsibility remains clear to citizens.

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A Additional Figures

A.1 Administrative organisation

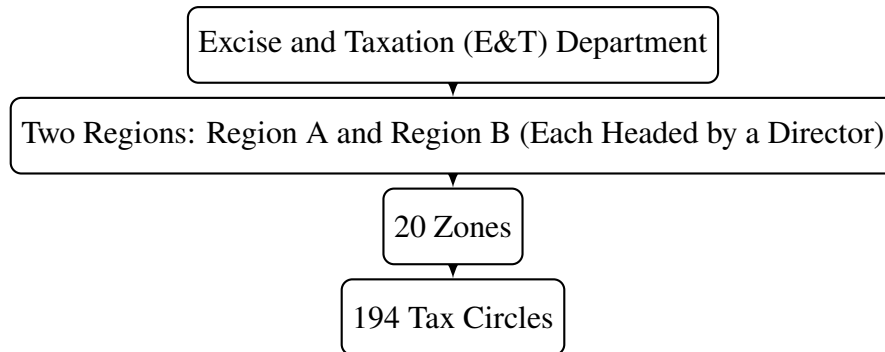


Figure A.1: Flowchart of Property Tax Administration in Lahore

A.2 Spatial coverage for survey sampling frame

RS estimation sample and E&T rating jurisdiction, Lahore district

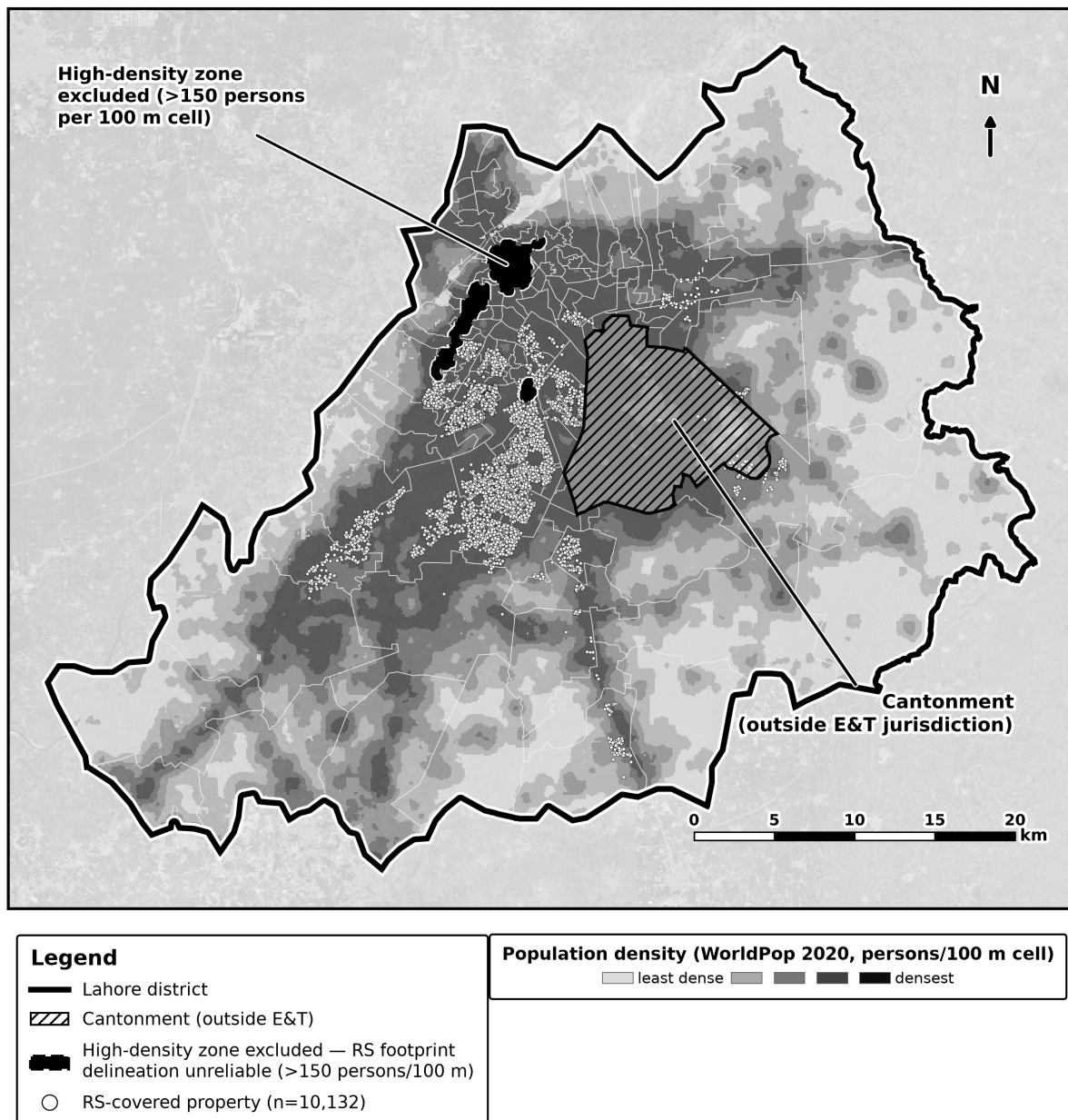


Figure A.2: The remote-sensing estimation sample within Lahore district. Grayscale shading is the WorldPop 2020 population surface (denser cells darker); thin lines are E&T rating jurisdiction. The Lahore Cantonment (hatched) is rated by the Cantonment Board and lies outside E&T jurisdiction; the dashed zone marks the high-density old-city fabric (above 150 persons per 100 m cell) where footprint delineation from imagery is unreliable. Open circles are the $n = 10,564$ RS-covered properties.

B Payoff comparisons and acceptance conditions

B.1 Utility comparisons

Write H_0 for an inspector alternative without a bribe, H_1 for a paid inspector alternative, and S for an automated alternative. For inspector alternatives at the same initial state e , define the net saving $s = p_{H_0} - p_{H_1}$. Suppressing the owner subscript, equation (1) gives

$$\begin{aligned} V_{H_1} - V_{H_0} &= [-p_{H_1} + h(e) - m(e)] - [-p_{H_0} + h(e)] \\ &= s - m(e). \end{aligned}$$

Thus $V_{H_1} \geq V_{H_0}$ if and only if $s \geq m(e)$, proving equation (2), including zero and negative $m(e)$.

For alternatives without bribes with common payment and state,

$$V_S - V_{H_0} = [-p + h(e) + \beta(e)] - [-p + h(e)] = \beta(e).$$

Comparing this utility difference at e and 0 gives $\beta(e) - \beta(0)$. Direct state valuation cancels within each comparison.

Likewise, define the paid-minus-bribe-free inspector contrast at common expenditure as $Q(e) = V_{H_1}(p, e) - V_{H_0}(p, e)$. Then

$$Q(e) = -m(e), \quad Q(e) - Q(0) = -[m(e) - m(0)].$$

This isolates state dependence in the paid-transaction term under the model. By contrast, two paid inspector alternatives at equal expenditure give

$$V_{H_1}(p, e) - V_{H_1}(p, 0) = h(e) - [m(e) - m(0)],$$

using $h(0) = 0$. The general error response remains. Equal bribe amounts do not remove it.

Finally, compare a paid inspector alternative at state e_H with automation at state e_S . Defining $s = p_S - p_H$,

$$\begin{aligned} V_H - V_S &= [-p_H + h(e_H) - m(e_H)] - [-p_S + h(e_S) + \beta(e_S)] \\ &= s + h(e_H) - h(e_S) - m(e_H) - \beta(e_S). \end{aligned}$$

The inspector offer is weakly preferred precisely when

$$s \geq \beta(e_S) + m(e_H) + h(e_S) - h(e_H).$$

At a common state this reduces to $\beta(e) + m(e)$. Omitting the state-valuation difference when states differ requires an additional restriction.

C Survey questions and coding

The following extract retains the questions required to interpret the reported survey measures. Each non-BACE outcome has its own coding and denominator. Codes 97 (don't know) and 98 (refusal), blanks and other non-substantive values are excluded from substantive-response percentages and shown separately where relevant. The 2,100 completed owner interviews are distinct from the contextual Phase 1 citizen and bureaucrat surveys.

Table C.1: Questionnaire items for the outcomes and respondent characteristics

Measure and item	Question	Responses and routing
Expected correctability S5.5	If a computer assessment of your property were wrong, how easy to get it corrected?	1 = Very hard; 2 = Hard; 3 = Easy; 4 = Very easy. Asked before the choice tasks, after earlier survey material.
Evidence demand S6.1	You can receive the computer's estimated details for your own house (covered area, plot outline). The price would be taken from your Rs 1,000 survey payment. Would you like to buy this for Rs {0/10/20/50}? (Rs 0 = free)	Yes; No; Don't know/undecided; Refused. The archived prices in the question differ from the analysed Rs 0, Rs 100 and Rs 200 offers; see notes.
Correction intention S9.1	In the next 3 months, how likely are you to apply to check or correct this house's tax record – out of 10?	0–10. Non-substantive responses use separate side codes.
Reason for inaction S9.2	If you do NOT plan to contest, what is the main reason?	1 = Assessment seems fair; 2 = Too costly to appeal; 3 = Don't know how to appeal; 4 = Afraid of consequences; 5 = Too much hassle; 77 = Other (specify). Asked when intention is 0–4.
Digital comfort S1.6	How comfortable are you with filling a form online?	1 = Not at all comfortable; 2 = A little comfortable; 3 = Comfortable; 4 = Very comfortable.
Document access S3.17	When you need them for property or other official work, how easy or difficult is it to find these documents for this property?	1 = Very difficult; 2 = Difficult; 3 = Somewhat difficult; 4 = Easy; 5 = Very easy.

Table C.1: Questionnaire items for the outcomes and respondent characteristics (continued)

Measure and item	Question	Responses and routing
Available documents S3.16	For this property, which of these do you currently have AND could produce within about a week? (read each)	Each item: 1 = Yes, have & can produce; 0 = No / cannot. Items: registry/sale deed; architectural drawings recording official land area and covered area; latest property-tax receipt/PT-1.
Reported title S3.14	In official records, how is this property currently held?	1 = Solely in my name; 2 = Jointly owned / shared title; 3 = In another (family) member's name (e.g. inherited; transfer/intiqal not done); 4 = Ownership is disputed / under litigation; 5 = No formal ownership record exists.
Officer trust S3.9	In general, how much do you trust property-tax officers to value properties fairly?	1 = Not at all; 2 = A little; 3 = Mostly; 4 = Completely.
Recent reassessment S3.6	In last six months, have you received any communication from E&T about change in assessment either informally or formally through a letter	1 = Yes; 0 = No.
Notice accuracy S3.7	If yes, is the new assessed value accurate?	1 = Much too low; 2 = Somewhat too low; 3 = About right; 4 = Somewhat too high; 5 = Much too high. Asked following a reported reassessment notice.

Notes: Questions and substantive categories reproduce the English extracts documented in the original manuscript, with typographic standardisation. Except for the evidence offer, the listed items additionally allow 97 = Don't know and 98 = Refused (or Refused to answer). These responses remain outside the substantive scales.

The document question's Urdu parenthetical for the sale deed is omitted here. The document count requires substantive answers to all three items; non-substantive answers are not treated as absence. Reported title is not verified legal eligibility. Correctability combines Easy and Very easy in the analysis and is a pre-choice belief, not an assigned treatment or a measure established to precede every survey exposure. The evidence-offer wording is from the archived instrument: its price list is not the deployed price support analysed in this paper. Acceptance is distinguished from verified payment, delivery and use in a successful correction. The analysis codes evidence acceptance as one and rejection as zero; other responses are excluded. The archived English S3.6 wording specifies six months, whereas its Urdu version specifies one year; the text therefore refers to a recent communication. The archived instrument does not establish the evidence offer's final placement.

Table C.2: Intention to seek correction after an assessment judged too high

Response	Owners	Percent
0	1	0.6
1	0	0.0
2	0	0.0
3	6	3.6
4	38	22.5
5	93	55.0
6	21	12.4
7	6	3.6
8	3	1.8
9	1	0.6
10	0	0.0
Don't know (97)	0	0.0
Refused (98)	0	0.0
Missing/other non-substantive	0	0.0
Sample total	169	100.0
Substantive answers	169	100.0

Notes: Sample: 169 owners reporting recent assessment communication and judging the value somewhat or much too high. The question asks the likelihood, out of ten, of applying to check or correct the house's tax record in the next three months. Low intention is 0–4. This is intention, not an observed application or failed correction. Percentages use the full stated sample, including non-substantive answers. Unknown, refused and missing/other responses are excluded from valid-answer summaries.

Table C.3: Reasons for low intention to seek correction

Response	Owners	Percent
Assessment seems fair	2	4.4
Too costly to appeal	0	0.0
Do not know how to appeal	0	0.0
Afraid of consequences	16	35.6
Too much hassle	27	60.0
Other reason	0	0.0
Don't know (97)	0	0.0
Refused (98)	0	0.0
Missing/other non-substantive	0	0.0
Sample total	45	100.0
Substantive answers	45	100.0

Notes: Sample: the 45 owners with intention 0–4 among recent notice recipients judging the assessment too high. The question asks the main reason for not planning to contest. Fear and hassle are separate reported reasons, not verified barriers to a submitted application. Percentages use the full stated sample, including non-substantive answers. Unknown, refused and missing/other responses are excluded from valid-answer summaries.

Table C.4: Comfort with completing an online form

Response	Owners	Percent
Not at all comfortable	85	4.0
A little comfortable	669	31.9
Comfortable	665	31.7
Very comfortable	662	31.5
Don't know (97)	19	0.9
Refused (98)	0	0.0
Missing/other non-substantive	0	0.0
Sample total	2,100	100.0
Substantive answers	2,081	99.1

Notes: Sample: all 2,100 completed owner interviews. Low comfort combines not at all comfortable and a little comfortable (codes 1–2). This is self-reported comfort, not a test of digital skills. Percentages use the full stated sample, including non-substantive answers. Unknown, refused and missing/other responses are excluded from valid-answer summaries.

Table C.5: Reported difficulty finding property documents

Response	Owners	Percent
Very difficult	388	18.5
Difficult	493	23.5
Somewhat difficult	462	22.0
Easy	597	28.4
Very easy	130	6.2
Don't know (97)	1	0.0
Refused (98)	29	1.4
Missing/other non-substantive	0	0.0
Sample total	2,100	100.0
Substantive answers	2,070	98.6

Notes: Sample: all 2,100 completed owner interviews. Difficulty includes very difficult, difficult and somewhat difficult (codes 1–3). This is distinct from possession of the three documents. Percentages use the full stated sample, including non-substantive answers. Unknown, refused and missing/other responses are excluded from valid-answer summaries.

D BACE conjoint: respondent-facing scenarios

Figures D.1–D.5 reproduce the five original illustrations supplied with the design documentation: an example receipt pair and the four common opening scenarios. The images retain their original wording, option order and displayed amounts. The Rs 20,000 reference is illustrative; it is not a common liability assigned to every owner. Outcome definitions and estimation are in Sections 5 and 6.2. Table D.1 records the realised eligible samples, which differ from these illustrative screens.

Below are two possible assessment situations for your own house. They differ in who assesses the house, how accurate it is, how much lower than the correct amount the recorded bill is, and any informal payment involved.

Please choose the situation you would prefer for your own house.

	OPTION A	OPTION B
Your real bill	Rs 20,000	Rs 20,000
Who checks your house	 An inspector visits & measures	 A satellite checks automatically
Measuring mistake (this time)	+ Rs 1,000 recorded too HIGH (by mistake)	Correct amount (no mistake)
Bill after the mistake	Rs 21,000	Rs 20,000
The deal (chai-pani)	Pay Rs 5,000 and the inspector records Rs 11,900 instead of Rs 21,000	No deal, no one to bargain with
Informal payment	Rs 5,000	Rs 0
Final recorded bill	Rs 11,900	Rs 20,000
What actually leaves your pocket	Rs 11,900 + Rs 5,000 Rs 16,900	Rs 20,000

Choose A

Choose B

Figure D.1: Illustrative BACE receipt pair

Notes: Original illustration supplied with the questionnaire documentation. It shows how assessor, initial bill, informal payment, final official bill and total expenditure are presented. The illustrative amounts do not imply a common reference liability for every owner.

	OPTION A	OPTION B
Your real bill	Rs 20,000	Rs 20,000
Who checks your house	 A satellite checks automatically	 An inspector visits & measures
Measuring mistake (this time)	Correct amount (no mistake)	Correct amount (no mistake)
Bill after the mistake	Rs 20,000	Rs 20,000
The deal (chai-pani)	No deal, no one to bargain with	No payment offered
Informal payment	Rs 0	Rs 0
Final recorded bill	Rs 20,000	Rs 20,000
What actually leaves your pocket	Rs 20,000	Rs 20,000

Choose A

Choose B

Figure D.2: Scenario 1: satellite and inspector at equal correct bills

	OPTION A	OPTION B
Your real bill	Rs 20,000	Rs 20,000
Who checks your house	 An inspector visits & measures	 An inspector visits & measures
Measuring mistake (this time)	Correct amount (no mistake)	Correct amount (no mistake)
Bill after the mistake	Rs 20,000	Rs 20,000
The deal (chai-pani)	No payment offered	Pay Rs 3,000 and the inspector records Rs 14,600 instead of Rs 20,000
Informal payment	Rs 0	Rs 3,000
Final recorded bill	Rs 20,000	Rs 14,600
What actually leaves your pocket	Rs 20,000	Rs 14,600 + Rs 3,000 Rs 17,600

Choose A

Choose B

Figure D.3: Scenario 2: inspector offers with and without a paid concession

	OPTION A	OPTION B
Your real bill	Rs 20,000	Rs 20,000
Who checks your house	 An inspector visits & measures	 An inspector visits & measures
Measuring mistake (this time)	Correct amount (no mistake)	+ Rs 3,000 recorded too HIGH (by mistake)
Bill after the mistake	Rs 20,000	Rs 23,000
The deal (chai-pani)	Pay Rs 2,000 and the inspector records Rs 17,000 instead of Rs 20,000	Pay Rs 4,000 and the inspector records Rs 15,000 instead of Rs 23,000
Informal payment	Rs 2,000	Rs 4,000
Final recorded bill	Rs 17,000	Rs 15,000
What actually leaves your pocket	Rs 17,000 + Rs 2,000 Rs 19,000	Rs 15,000 + Rs 4,000 Rs 19,000

Choose A

Choose B

Figure D.4: Scenario 3: different initial assessment states at equal total expenditure

	OPTION A	OPTION B
Your real bill	Rs 20,000	Rs 20,000
Who checks your house	 An inspector visits & measures	 An inspector visits & measures
Measuring mistake (this time)	Correct amount (no mistake)	+ Rs 2,000 recorded too HIGH (by mistake)
Bill after the mistake	Rs 20,000	Rs 22,000
The deal (chai-pani)	Pay Rs 3,000 and the inspector records Rs 16,000 instead of Rs 20,000	Pay Rs 3,000 and the inspector records Rs 16,000 instead of Rs 22,000
Informal payment	Rs 3,000	Rs 3,000
Final recorded bill	Rs 16,000	Rs 16,000
What actually leaves your pocket	Rs 16,000 + Rs 3,000 Rs 19,000	Rs 16,000 + Rs 3,000 Rs 19,000

Choose A

Choose B

Figure D.5: Scenario 4: different initial assessment states at equal expenditure and informal payment

Table D.1 applies each comparison's restrictions to the displayed receipt amounts. Failing an equality condition excludes a task from that common-comparison share, not automatically from full-sequence estimation. The reference correct bill is part of the hypothetical exercise, not an independent adjudication of legal liability.

Table D.1: Common comparisons satisfying the displayed conditions

Choice	Chosen / eligible	Percent	95% interval
Satellite: correct bill, equal expenditure, no deal	1,256/2,100	59.8	[57.7, 61.9]
Inspector deal: correct bills, deal versus no deal	1,122/2,050	54.7	[52.6, 56.9]
Overassessment-starting deal: equal expenditure	219/927	23.6	[21.0, 26.5]
Overassessment-starting deal: equal expenditure and payment	423/1,624	26.0	[24.0, 28.2]

Notes: Each initial task has 2,100 choices. Eligibility imposes the stated conditions on the receipts. Both sides of tasks 2–4 are inspectors; tasks 3–4 contain payments on both sides. Median net savings in the qualifying second task are Rs 1,200. Wilson intervals describe choices as presented, without removing position or order effects.

E Sample description, likelihood and sensitivity

Table E.1: Recorded property size by survey completion

Attribute	Completed	Not completed
Administrative land area, FY2026–27 (marla)	15.3 / 10.0 (2,024)	15.8 / 10.0 (3,831)
Remote-sensing land area (marla)	17.0 / 9.4 (2,073)	17.4 / 8.5 (3,938)

Notes: Entries are mean / median (valid observations), using the existing administrative panel and remote-sensing benchmark linked by property identifier. All 6,088 survey-frame properties match the panel. Non-positive and non-numeric values are excluded. This comparison describes selection on recorded size; it cannot establish equivalence in unobserved trust or preferences.

The administrative comparison links unique property identifiers rather than treating each annual record as a separate property. The survey file contains 6,088 unique properties, all of which match the linked administrative panel; 2,100 have a completed interview. The size comparison uses positive observed values and reports the denominator separately for each source and participation group. The small preloaded administrative-information fields in v25 are not used for this comparison because they cover only a subset of respondents.

The survey’s trust and corruption items distinguish institutional perceptions from personal experience. Among valid responses, 1,054 of 2,072 owners report little or no trust that officers value properties fairly. Of 2,034 answering the general informal-payment prevalence question, 102 report often or very often, and 535 report sometimes or more frequently. Among 1,188 contacted owners giving a valid personal-experience answer, 159 report being asked for an informal payment. These different quantities neither replicate the baseline collection-gap attribution question nor form a time series with it.

Table E.2: Support across the complete choice sequence

Assessor	Initial bill	No paid deal	Paid deal
Inspector	Correct	4,735	7,754
Inspector	Overassessed	886	10,815
Inspector	Underassessed	1,097	3,702
Satellite	Correct	4,071	0
Satellite	Overassessed	3,869	0
Satellite	Underassessed	5,073	0

Notes: Counts are offered alternatives, not respondents or chosen offers. The 42,002 alternatives form 21,001 paired tasks from 2,100 owners. Satellite offers never include a paid deal. The analysis uses displayed rupee amounts and preserves each original task slot.

Receipt attributes are reconstructed from the amounts shown. Error equals initial minus correct liability; the official reduction equals initial minus final official liability; expenditure equals

the final official liability plus the requested payment. Every retained receipt satisfies these identities and each parsed paid-offer sentence agrees with the stated amounts. Four alternatives show a Rs 100 reduction without an informal payment; the paid-deal indicator requires a strictly positive payment. Every respondent supplies ten valid tasks. One also supplies a valid slot-12 task; the intervening slot has no usable choice. Omitting the additional task leaves the substantive results unchanged.

For two alternatives, conditioning on the task eliminates any term common to the offers. Defining $\Delta X_{it} = X_{iBt} - X_{iAt}$ gives the binary likelihood in (5). The Stata conditional-logit implementation and an independent maximum-likelihood implementation on these differences agree on the primary coefficients to within 10^{-5} . The design matrix of within-task differences has full column rank for each retained specification. That algebraic condition permits estimation under the model; it is not a guarantee of broad experimental support for every counterfactual.

The mixed specification sets $\alpha_i = \bar{\alpha} + \sigma_\alpha z_{\alpha i}$ and $\kappa_i = \bar{\kappa} + \sigma_\kappa z_{\kappa i}$, where the two z terms are independent standard normal variables. The expenditure coefficient remains common. The panel mixed likelihood is

$$L_i(\vartheta) = \int \int \prod_{t=1}^{T_i} p_{it}(z; \vartheta)^{y_{it}} [1 - p_{it}(z; \vartheta)]^{1-y_{it}} \phi(z_\alpha) \phi(z_\kappa) dz_\alpha dz_\kappa.$$

The same draws affect all of an owner's choices, so integration follows the product over tasks. Integrating task by task would fail to represent persistent preferences. Estimation uses Gauss–Hermite quadrature with 17 and 25 nodes per dimension. The uncertainty calculation uses the inverse numerical Hessian around the optimum and the outer products of respondent likelihood scores.

For a sequential assignment rule depending on recorded history $H_{i,t-1}$, the joint likelihood also contains factors for assignment. These factors can be omitted for preference estimation only when they carry no additional dependence on the unknown preference parameters conditional on that history. Dependence on unrecorded current shocks would violate the maintained interpretation. The available local implementation demonstrates a link to previous answers but does not fully establish the complete rule. The results are consequently conditional preference estimates, with a homogeneous model and a specified random-effects model reported together.

E.1 Model restrictions and conditional comparisons

Tables 5 and E.3 connect the estimated equation to the restrictions in Section 3. For a vector of restrictions $Rb = 0$, the Wald statistic is

$$W = (Rb)'(R\widehat{\Sigma}R')^{-1}(Rb),$$

with a chi-squared reference distribution whose degrees of freedom equal the number of independent restrictions. A linear comparison $a'b$ has standard error $\sqrt{a'\widehat{\Sigma}a}$; the table reports

normal-reference 95% intervals. Here b collects the coefficients and $\widehat{\Sigma}$ is their covariance matrix, using respondent clustering or respondent likelihood scores. The calculations use the saved full-precision estimates and covariance matrices corresponding to the reported models. The reported tests are two-sided and unadjusted for multiple comparisons.

In the mixed model, κ denotes the mean paid-deal coefficient. The joint zero-coefficient test therefore concerns mean transaction valuation, not the additional restriction of zero heterogeneity required for savings alone for every owner. State interactions are common across owners in this specification. The contrast $\lambda_O - \lambda_U$ asks whether the change in transaction valuation differs between overassessment and underassessment; a positive λ_O alone does not establish that difference.

Table E.3: Conditional state and arrangement comparisons

Comparison	Pooled logit	Panel mixed logit	Later tasks
Unpaid human: over versus correct	-0.625 [-0.799, -0.450]	-0.971 [-1.184, -0.758]	-0.245 [-0.544, 0.054]
Unpaid human: under versus correct	-0.588 [-0.759, -0.417]	-0.489 [-0.689, -0.289]	-0.359 [-0.648, -0.069]
Paid human: over versus correct	-0.575 [-0.634, -0.516]	-0.682 [-0.749, -0.616]	-0.099 [-0.224, 0.026]
Paid human: under versus correct	-0.581 [-0.697, -0.465]	-0.208 [-0.343, -0.073]	-0.285 [-0.425, -0.146]
Automated: over versus correct	-0.531 [-0.659, -0.404]	-1.085 [-1.249, -0.922]	-0.310 [-0.454, -0.165]
Automated: under versus correct	-0.446 [-0.572, -0.320]	-0.377 [-0.531, -0.222]	-0.357 [-0.495, -0.219]
Paid versus unpaid human, correct	0.381 [0.290, 0.472]	0.142 [0.018, 0.265]	0.316 [0.042, 0.591]
Paid versus unpaid human, over	0.431 [0.274, 0.588]	0.431 [0.234, 0.627]	0.462 [0.299, 0.626]
Paid versus unpaid human, under	0.388 [0.242, 0.534]	0.423 [0.244, 0.601]	0.390 [0.242, 0.538]
Automated versus unpaid human, correct	0.056 [-0.036, 0.148]	0.074 [-0.050, 0.197]	0.221 [-0.046, 0.487]
Automated versus unpaid human, over	0.149 [-0.011, 0.309]	-0.041 [-0.249, 0.168]	0.156 [-0.005, 0.317]
Automated versus unpaid human, under	0.198 [0.052, 0.344]	0.186 [0.000, 0.371]	0.222 [0.077, 0.367]

Notes: Entries are systematic-utility differences in logit units, with 95% normal-reference confidence intervals in brackets, holding total expenditure and position fixed. Error comparisons are γ_e for unpaid human assessment, $\gamma_e + \lambda_e$ for paid human assessment and $\gamma_e + \delta_e$ for automation. Paid-versus-unpaid comparisons are κ at a correct bill and $\kappa + \lambda_e$ otherwise; assessor comparisons are α and $\alpha + \delta_e$. Intervals use the full covariance matrix and respondent clustering or respondent likelihood scores. Mixed-model arrangement comparisons use the mean random coefficient. Estimates are conditional on the task design and model; they do not identify a unique psychological mechanism. Intervals are unadjusted for multiple comparisons.

Table E.4: Sensitivity of error interactions and financial responses

Specification	Satellite $\times$ over	Satellite $\times$ under	Cash term
Full sequence	0.093 (0.088)	0.142* (0.082)	-0.097*** (0.017)
Later tasks	-0.065 (0.159)	0.001 (0.154)	-0.115*** (0.022)
Omit extra task	0.093 (0.088)	0.145* (0.082)	-0.097*** (0.017)
Omit option position	-0.007 (0.087)	0.045 (0.081)	-0.088*** (0.016)
Add requested payment	0.085 (0.088)	0.136* (0.082)	0.028*** (0.007)
Rupee error magnitudes	-0.050 (0.039)	-0.024 (0.024)	-0.077*** (0.016)

Notes: Standard errors are respondent clustered. In the error-magnitude row, interactions are per Rs 1,000 of error; other rows use error indicators. The final column is total expenditure per Rs 1,000, except in the additional-payment row, which reports the requested payment conditional on total expenditure. In the separate-component model (not tabulated), the payment coefficient is -0.032 (0.029) and the official-reduction coefficient is 0.065 (0.021), both per Rs 1,000. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$; two-sided normal-reference Wald tests using the reported standard errors. Stars refer to individual coefficients, not joint tests.

The sensitivity table separates alternative expenditure and error specifications instead of treating all coefficients as the same estimand. Magnitude specifications measure effects per Rs 1,000 of error; state-indicator specifications compare an error state with a correct bill. The separate-component model uses payment and reduction without total expenditure. The additional-payment model conditions on the same total, so changing its payment requires an offsetting change elsewhere in the receipt.

The baseline payoff function has no additional dependence on the size of a positive bribe conditional on expenditure and state. The added requested-payment coefficient is 0.028 (standard error 0.007 ; two-sided $p < 0.001$), providing evidence against that exclusion within the augmented specification. The response can concern the composition or presentation of expenditure; it does not identify a unique mechanism. Allowing $m_i(e, B)$ would relax the exclusion, while the baseline retains $m_i(e)$ to organise the main conditional comparisons.

Table E.5: Correctability beliefs and full-sequence assessor choices

Interaction	Beliefs	Add trust	Add arm interactions
Satellite $\times$ hard	0.534*** (0.071)	0.561*** (0.072)	0.565*** (0.072)
Satellite $\times$ easy/very easy	0.550*** (0.079)	0.652*** (0.082)	0.648*** (0.083)
Satellite $\times$ low officer trust	—	0.323*** (0.062)	0.324*** (0.063)
Respondents	2,032	2,004	2,004
Tasks	20,320	20,040	20,040

Notes: All specifications include the terms in equation (4). Very hard is the reference correctability category. Low officer trust combines not at all and a little in the current survey. The third column allows all ten baseline choice coefficients to differ across the three recorded information arms. Correctability is measured after information scripts; these coefficients are descriptive associations. Standard errors are clustered by respondent. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$; two-sided normal-reference Wald tests using the reported standard errors. Stars refer to individual coefficients, not joint tests.

F Additional evidence results

Table F.1: Demand for independent evidence about the owner's property

Posted price	Accept / observed	Percent	95% interval
Rs 0	633/668	94.8	[92.8, 96.2]
Rs 100	284/728	39.0	[35.5, 42.6]
Rs 200	277/703	39.4	[35.9, 43.1]

Notes: All valid purchase responses, measured after the conjoint. One owner assigned Rs 200 has no valid response. Intervals are Wilson binomial intervals. Accepting the offer records demand for the evidence, not a subsequent correction.

Table F.2: Posted prices and purchase decisions

Term	Percentage points (SE)	95% interval
Free offer	94.76*** (0.86)	[93.07, 96.45]
Rs 100 versus free	-55.75*** (2.01)	[-59.68, -51.82]
Rs 200 versus free	-55.36*** (2.04)	[-59.35, -51.36]

Notes: Saturated linear probability model, 2,099 observed responses. HC3 standard errors and two-sided t-reference inference. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The free-offer constant is the purchase share; other coefficients compare each paid offer with free provision.

The posted-price comparison retains 668 valid responses at zero, 728 at Rs 100 and 703 at Rs 200. One additional owner in the Rs 200 group supplies no substantive purchase answer. The saturated model reports each paid-price contrast separately. It does not require willingness to pay to follow a specified distribution, and the lack of a further decline between the two paid offers is visible in the reported shares.